Exact arithmetic for the Suzuki classification table

Human Compression Project

Abstract

The previously proved classification formulas reduce each numerical row to three integer factorizations. We acquire certificates for all 180 integers occurring at the intended exponents $1 \le e \le 60$, and substitute them in the accepted formulas. The numerical conclusions retain the original exact values and inequalities separately from their corrected values.

1 The arithmetic certificate

For $q = 2^{2e+1}$ the three integers are

$$q - 1, \qquad q + 2^{e+1} + 1, \qquad q - 2^{e+1} + 1.$$

The classification formulas use their numbers of distinct prime divisors. Factoring software supplies candidate factorizations; acceptance uses only integer multiplication, modular exponentiation and greatest common divisors.

Lemma 1. *Let $N > 2$, and suppose the complete prime factorization of $N - 1$ has been certified. If an integer a satisfies*

$$a^{N-1} \equiv 1 \pmod{N}, \qquad \gcd\bigl(a^{(N-1)/\ell} - 1, N\bigr) = 1 \quad \textit{for every prime } \ell \mid N - 1,$$

then N is prime.

Proof. Choose a prime divisor r of N. The first congruence makes a invertible modulo r and makes its order divide $N - 1$. For each prime $\ell \mid N - 1$, the greatest-common-divisor condition shows that this order does not divide $(N-1)/\ell$. Its ℓ-part is therefore the entire ℓ-part of $N - 1$. Hence the order is $N - 1$. Since it also divides $r - 1$, we have $r \ge N$. As $r \mid N$, equality holds and N is prime. □

This is the classical elementary criterion based on the factorization of $N-1$; no new primality theorem is claimed. Starting at 2, recursive certificates establish every prime used in the three factorizations. Multiplying the certified prime powers then proves equality with the target integer. Counting the distinct certified primes gives the required input to the classification formulas. No probable-prime answer is an acceptance premise. The PARI/GP factorization routine is used solely to propose missing factors; the integer reader is independent of its factoring algorithms.

2 The complete substitution

The final certificate joins the 134 previously accepted target factorizations with the 46 remaining targets. Their domains are disjoint and their union is exactly the three displayed integers for each

intended exponent $e = 1, \ldots, 60$. The completed collection uses 694 certified primes: 417 retained certificates and 277 added certificates. It reproduces all 34 previously completed numerical rows before extending the table to 60.

Substitution into the already proved classification formulas gives the following dispositions for the source's total column:

printed exact totals recovered	18
printed lower bounds confirmed	11
printed exact totals corrected	30
printed lower bounds refuted	1.

At $e = 39$ the printed lower bound 10^{23} is false. The exact total is

$$191285730951681831440 89 < 10^{23}.$$

There are also two errors in the printed distinct-prime counts: the minus torus has three prime divisors at $e = 43$, rather than two, and four at the intended $e = 60$, rather than three. The last right-hand row is literally printed with exponent 30; its position and parameters correspond to 60. Both the printed exponent and this contextual correction are retained.

The independent integer acquisition and the complete row data are part of the accompanying certificate package. The comparison treats every printed lower bound as an inequality. It neither rounds a corrected exact total to force agreement nor replaces a printed glyph silently. The example totals mentioned in Section 9 at $e = 3, 6, 9$ are respectively

$$2411, \qquad 6453727, \qquad 18084183211,$$

so those particular examples and their stated bounds are recovered. The broader conclusions concerning proper subfields require the corrected central-cover classification and are addressed there.

References

[1] D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie **39** (1998), 97–120, especially Table 12 and Section 9.